

Frame characterizations for time-dependent source recovery

Natural ℓ^p and $W^{1,p}$ formulations of arXiv:2401.15450

Autonomous proof packet — agent_lane_18 (GPT5.6)

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Abstract

Let $C : \mathcal{H} \rightarrow \ell^2$ be the analysis operator of a Bessel family $\mathcal{G} = (g_j)$, and let $A \in \mathcal{B}(\mathcal{H})$. For every $1 \leq p \leq \infty$, stable recovery of all discrete time-dependent sources from

$$x_{n+1} = Ax_n + w_n, \quad y_n = Cx_n,$$

as a bounded map $\ell^p(\mathbb{N}_0; \ell^2) \rightarrow \ell^p(\mathbb{N}_0; \mathcal{H})$, is possible if and only if $\mathcal{G}$ is a frame. The recovery is $(Ry)_n = Ly_{n+1} - ALy_n$, where $LC = I$. The continuous bounded-generator analogue also holds: universal stable recovery $W^{1,p}(0, T; \ell^2) \rightarrow L^p(0, T; \mathcal{H})$ is possible if and only if $\mathcal{G}$ is a frame, with $R(y) = Ly' - ALy$. These theorems answer a standard natural formulation of the source's time-dependent future-work direction. They do not settle its broader unbounded-semigroup program.

1 Future-work statements in the source

Section 8.2 of Aldroubi–Díaz Martín–Gong–Mashreghi–Medri, arXiv:2401.15450v1, proposes two generalizations. The exact page-22 text is shown below.

$$\mathcal{R}(\mathcal{D}(x_0, w)) = \sum_{j=1}^{\infty} j \langle x_1 - x_0, \frac{1}{j} e_j \rangle e_j = \sum_{j=1}^{\infty} \langle w, e_j \rangle e_j = w.$$

In conclusion, although $\mathcal{R}$ is a linear map and allows source reconstruction, it is an unbounded operator.

8.2. Generalizations of the source recovery problem.

8.2.1. *Time-dependent source term.* If the source term depends on time, that is, if the dynamical system (1.1) is replaced by one of the form

$$x_{n+1} = Ax_n + w_n, \quad n \in \mathbb{N},$$

where for each $n \in \mathbb{N}$, $w_n \in \mathcal{H}$, the reconstruction formula provided by Theorem 3.2 can be adopted in this framework. Indeed, assume that the spatial sampling vectors $\mathcal{G} = \{g_j\}_{j \geq 1}$ form a frame for $\mathcal{H}$. Then, for recovering each source term w_n , two iterations of the dynamical system are needed: Given the measurements $\{\langle x_n, g_j \rangle, \langle x_{n+1}, g_j \rangle\}_{j \geq 1}$, the source term w_n can be linearly recovered by

$$(8.1) \quad w_n = \sum_{j=1}^{\infty} \left(\langle x_{n+1}, g_j \rangle - \sum_{i=1}^{\infty} \overline{a_{ij}} \langle x_n, g_i \rangle \right) \tilde{g}_j$$

where $\{\tilde{g}_j\}_{j \geq 1}$ is a dual frame for $\mathcal{G}$ and the coefficients a_{ij} are given according to formula (5.11) (i.e., $a_{ij} = \langle A^* g_j, \tilde{g}_i \rangle$) (cf. (5.13)). Notice that if the reconstruction formulas (8.1) are used, then we need all the measurements $\mathcal{D}(x_0, w) = [\langle x_n, g_j \rangle]_{n \geq 0, j \geq 1}$ for recovering the source terms $\{w_n\}_{n \in \mathbb{N}}$. The exploration of natural conditions on the measurements $\mathcal{D}(x_0, w)$, as well as necessary and sufficient on $\mathcal{G}$ to guarantee stable recovery in this case, are part of future work.

8.2.2. *Continuous-time dynamical systems.* As part of future work, we aim to explore a continuous version of the source recovery problem and address some similarities with the discrete version. Precisely, consider the continuous-time dynamical system on a separable Hilbert space $\mathcal{H}$

$$(8.2) \quad \begin{cases} \dot{x}(t) = Ax(t) + w, & t \in \mathbb{R}_{\geq 0}, \\ x(0) = x_0 \end{cases}$$

where A is the infinitesimal generator of a strongly continuous semigroup $T : \mathbb{R}_{\geq 0} \rightarrow \mathcal{B}(\mathcal{H})$.

The first target concludes:

Notice that if the reconstruction formulas (8.1) are used, then we need all the measurements $\mathcal{D}(x_0, w) = [\langle x_n, g_j \rangle]_{n \geq 0, j \geq 1}$ for recovering the source terms $\{w_n\}_{n \in \mathbb{N}}$. The exploration of natural conditions on the measurements $\mathcal{D}(x_0, w)$, as well as necessary and sufficient on $\mathcal{G}$ to guarantee stable recovery in this case, are part of future work.

The source then asks for a continuous-time analogue and concludes on page 23:

The exploration of natural spaces for the measurements $\mathcal{D}_c$, as well as necessary and sufficient on $\mathcal{G}$ to guarantee stable recovery in this case, are part of future work.

These are programs rather than fixed propositions: stability has no unique meaning until the data and source norms are specified. We use the standard Bochner sequence spaces in discrete time and the strong Sobolev observation space in continuous time.

2 Frame and recovery conventions

Let $\mathcal{H}$ be a separable Hilbert space and let $\mathcal{G} = (g_j)_{j \geq 1} \subset \mathcal{H}$ be Bessel. Its analysis operator is

$$C : \mathcal{H} \longrightarrow \ell^2, \quad Ch = (\langle h, g_j \rangle)_{j \geq 1}.$$

The family $\mathcal{G}$ is a frame precisely when C is bounded below:

$$\|Ch\|_{\ell^2} \geq a\|h\|_{\mathcal{H}} \quad (h \in \mathcal{H})$$

for some $a > 0$. Equivalently, C has a bounded left inverse $L : \ell^2 \rightarrow \mathcal{H}$, for example synthesis by the canonical dual frame.

For $1 \leq p \leq \infty$, a *stable universal discrete recovery* is a bounded linear map

$$R_p : \ell^p(\mathbb{N}_0; \ell^2) \longrightarrow \ell^p(\mathbb{N}_0; \mathcal{H})$$

such that $R_p((Cx_n)_n) = (w_n)_n$ for every trajectory $x_{n+1} = Ax_n + w_n$ for which both the displayed data and source belong to their indicated spaces. The definition includes all finite-support trajectories.

3 Exact discrete characterization

Theorem 1 (time-dependent discrete sources). *Let $A \in \mathcal{B}(\mathcal{H})$, let $\mathcal{G}$ be Bessel, and fix $1 \leq p \leq \infty$. A stable universal discrete recovery exists if and only if $\mathcal{G}$ is a frame.*

If $LC = I$, one recovery operator is

$$(R_p y)_n = Ly_{n+1} - ALy_n, \quad n \geq 0, \quad (1)$$

and it satisfies

$$\|R_p\| \leq \|L\|(1 + \|A\|).$$

Conversely, the existence of any such R_p implies

$$\|Ch\|_{\ell^2} \geq \|R_p\|^{-1}\|h\|_{\mathcal{H}} \quad (h \in \mathcal{H}).$$

Proof. Assume first that $\mathcal{G}$ is a frame and choose L with $LC = I$. The unilateral shift $Sy = (y_{n+1})_{n \geq 0}$ has norm at most one on every ℓ^p , including $p = \infty$. Applying L coordinatewise, formula (1) defines the bounded operator

$$R_p = LS - AL,$$

with the asserted norm bound. For trajectory data $y_n = Cx_n$, we have

$$(R_p y)_n = LCx_{n+1} - ALCx_n = x_{n+1} - Ax_n = w_n.$$

Conversely, suppose R_p is a stable universal recovery. Given any $h \in \mathcal{H}$, define the finite-support trajectory

$$x_0 = 0, \quad x_1 = h, \quad x_n = 0 \quad (n \geq 2).$$

It is generated by

$$w_0 = h, \quad w_1 = -Ah, \quad w_n = 0 \quad (n \geq 2).$$

Its observation sequence has the single nonzero entry $y_1 = Ch$. Therefore, for finite p ,

$$\|h\| \leq (\|h\|^p + \|Ah\|^p)^{1/p} = \|w\|_{\ell^p} = \|R_p y\|_{\ell^p} \leq \|R_p\| \|Ch\|_{\ell^2}.$$

For $p = \infty$, the same chain starts with $\|h\| \leq \max(\|h\|, \|Ah\|) = \|w\|_{\ell^\infty}$. Thus C is bounded below. Since $\mathcal{G}$ was Bessel, it is a frame. $\square$

Remark 1. *Arbitrary time dependence is decisive in the necessity proof: a one-step source pulse creates an arbitrary state h , and the next pulse cancels it. No spectral or stability assumption on A is needed.*

4 Continuous bounded-generator upgrade

Let $I = (0, T)$, $0 < T < \infty$, and use the norm

$$\|y\|_{W^{1,p}(I;\ell^2)} = \|y\|_{L^p(I;\ell^2)} + \|y'\|_{L^p(I;\ell^2)}.$$

A *stable universal continuous recovery* is a bounded linear map

$$R_p^c : W^{1,p}(I; \ell^2) \longrightarrow L^p(I; \mathcal{H})$$

that recovers w from $y = Cx$ for every strong solution

$$x' = Ax + w, \quad x \in W^{1,p}(I; \mathcal{H}), \quad w \in L^p(I; \mathcal{H}).$$

Theorem 2 (time-dependent continuous forcing). *Let $A \in \mathcal{B}(\mathcal{H})$, let $\mathcal{G}$ be Bessel, and let $1 \leq p \leq \infty$. A stable universal continuous recovery exists if and only if $\mathcal{G}$ is a frame.*

If $LC = I$, then

$$R_p^c(y) = Ly' - ALy \tag{2}$$

is such a recovery and $\|R_p^c\| \leq \|L\| \max(1, \|A\|)$ for the sum norm above. Conversely, for every $\lambda > \|A\|$,

$$\|Ch\|_{\ell^2} \geq \frac{\lambda - \|A\|}{\|R_p^c\|(1 + \lambda)} \|h\|_{\mathcal{H}}.$$

Proof. If C has a bounded left inverse L , boundedness of L implies that it commutes with Bochner weak differentiation. Thus $Ly = x$ and $Ly' = x'$ for observation data $y = Cx$. Formula (2) gives

$$R_p^c(y) = x' - Ax = w.$$

Moreover,

$$\|R_p^c(y)\|_{L^p} \leq \|L\| \|y'\|_{L^p} + \|A\| \|L\| \|y\|_{L^p} \leq \|L\| \max(1, \|A\|) \|y\|_{W^{1,p}}.$$

For necessity, assume a stable recovery exists. Fix $\lambda > \|A\|$ and $h \in \mathcal{H}$, and choose the strong trajectory

$$x(t) = e^{\lambda t} h, \quad w(t) = e^{\lambda t} (\lambda I - A)h, \quad y(t) = e^{\lambda t} Ch.$$

Writing $E_p = \|e^{\lambda(\cdot)}\|_{L^p(0,T)}$, with the usual interpretation for $p = \infty$, the reverse triangle inequality gives

$$\|w\|_{L^p} = E_p \|(\lambda I - A)h\| \geq E_p (\lambda - \|A\|) \|h\|.$$

On the other hand,

$$\|y\|_{W^{1,p}} = E_p (1 + \lambda) \|Ch\|_{\ell^2}.$$

Since $R_p^c(y) = w$, boundedness of R_p^c and cancellation of $E_p > 0$ give the displayed lower bound for C . Hence $\mathcal{G}$ is a frame. $\square$

5 Boundary of the result

The source's continuous future direction allows an unbounded infinitesimal generator and an arbitrary $x_0 \in \mathcal{H}$, then informally uses $\mathcal{D}'_c(0)$. This derivative need not exist. For example, let $T(t)$ be the translation semigroup on $L^2(\mathbb{R})$, whose generator has domain $H^1(\mathbb{R})$, choose $x_0 \notin H^1(\mathbb{R})$, and let $\mathcal{G}$ be an orthonormal basis. Its analysis map is unitary, so the ℓ^2 -valued observation curve is strongly differentiable at zero if and only if $T(t)x_0$ is; it is not for this x_0 .

Thus a full unbounded-generator theory must restrict initial states, use weak or mild derivatives, or adopt an integral observation space. Also, the source’s continuous model has a constant w , whereas Theorem 2 requires recovery for arbitrary L^p forcing. Time evolution can create additional observability for a constant source, so the frame condition need not be necessary in every weaker formulation.

6 Novelty and review status

Local and web searches found arXiv:2401.15450, the authors’ earlier constant-source paper, and continuous dynamical-sampling work for initial-state recovery, but no later source containing Theorems 1–2. This is not proof of novelty. The discrete result substantially resolves the first future direction after a standard choice of Banach spaces; the continuous result resolves a bounded strong-solution analogue. Because the source asks a family of formulations rather than a single theorem, this packet is classified as a promoted partial result and should receive expert review.

Human-review checklist

Item	Check
Discrete definition	Confirm that universal recovery includes all finite-support trajectories.
Pulse necessity	Verify $w_0 = h, w_1 = -Ah$ and the one-row data norm.
Continuous definition	Confirm the chosen $W^{1,p}$ sum norm and strong-solution class.
Exponential test	Recheck the lower bound for $\lambda I - A$.
Scope	Do not infer the same necessity for constant forcing and arbitrary semigroups.
Novelty	Search journal versions and citing literature.

References

1. A. Aldroubi, R. Díaz Martín, L. Gong, J. Mashreghi, and I. Medri, *Characterization of frames for source recovery from dynamical samples*, arXiv:2401.15450v1 (2024).
2. A. Aldroubi, R. Díaz Martín, and I. Medri, *Dynamical Sampling for the Recovery of Spatially Constant Source Terms in Dynamical Systems*, arXiv:2308.01462 (2023).
3. R. Díaz Martín, I. Medri, and U. Molter, *Continuous and discrete dynamical sampling*, J. Math. Anal. Appl. 499 (2021), Paper No. 125060.